

StoryProjectionOnto: Conference-Study Results Report

Study status: **INCOMPLETE** — The primary 14B pilot was rejected; fallback acceptance, development execution, independent review, held-out runs, and the lawful first-novel study remain incomplete.

This report is generated from hash-verified canonical tables. Missing results are marked incomplete; no values are imputed.

Study overview

Narrative evidence does not determine one universally useful ontology. This study tests whether revealing a user's context before ontology construction changes the semantic fidelity and organization of the resulting evidence-grounded graph.

RQ1: Does query-dependent LLM construction improve qualified-assertion and ontology-decision fidelity relative to query-blind LLM and classical construction?

RQ2: How does construction timing change entropy and directly measured visual clutter when semantic and rare-pivotal safeguards remain visible?

RQ3: How does construction timing change gold-aligned community structure and cross-seed cluster stability?

The query-blind evidence index stores only source-grounded evidence and retrieval metadata; it is not a hidden ontology. C2 forms contextual entities, events, schema, relations, abstractions, and qualified assertions after query reveal.

1. Research questions and hypotheses

STATUS: INCOMPLETE — Contracts, resource controls, and GPU pilot: The 14B candidate was rejected and fallback acceptance has not completed.

No confirmatory or descriptive outcome is available for this section.

H1-semantic registers strict qualified-assertion F1 and H1-organization registers ontology-decision macro F1 for C2 versus C1. Rare-pivotal recall is a noncompensable safeguard, not another optimization target.

2. Experimental conditions and construction timing

STATUS: INCOMPLETE — Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.

No confirmatory or descriptive outcome is available for this section.

C0 and C1 construct before query reveal; C2 actively constructs after query reveal; A-FixedSelect uses query-time LLM selection over the complete sealed C1 ontology while construction operations are mechanically forbidden.

3. Temporal qualified-assertion representation

STATUS: INCOMPLETE — Contracts, resource controls, and GPU pilot: The 14B candidate was rejected and fallback acceptance has not completed.

No confirmatory or descriptive outcome is available for this section.

Assertions separate story time, validity time, discourse order, proposition revelation position, spoiler horizon, holder-level epistemic status, confidence, provenance, contextual relevance, and supported why-it-matters text.

4. Synthetic benchmark construction and split

STATUS: INCOMPLETE — Synthetic benchmark: The mandatory independent condition-blind review has not been supplied or adjudicated.

No confirmatory or descriptive outcome is available for this section.

The registered design contains four development worlds and twelve held-out worlds with three same-evidence contexts per held-out world. Held-out execution remains gated on independent condition-blind review.

5. Resource and reproducibility controls

STATUS: INCOMPLETE — Contracts, resource controls, and GPU pilot: The 14B candidate was rejected and fallback acceptance has not completed.; Results and conference artifacts: The reporting substrate exists, but final immutable result inputs do not.

No confirmatory or descriptive outcome is available for this section.

GPU time includes loading, warm-up, failures, repairs, restarts, and inference. Occupied project storage targets at most 25 GB with 5 GB protected headroom.

Interim verified table. It is not a completed section outcome.

Machine-readable phase status at this build

phase_id	status	reason
phase_1	incomplete	The 14B candidate was rejected and fallback acceptance has not completed.
phase_2	incomplete	The mandatory independent condition-blind review has not been supplied or adjudicated.
phase_3	incomplete	Development and held-out GPU condition outputs are not complete.
phase_4	incomplete	No registered held-out condition table exists to analyze.
phase_5	incomplete	GPU-backed scripted feedback and researcher traces have not executed.
phase_6	incomplete	No lawful local novel path has been supplied and no case-study output exists.
phase_7	incomplete	The reporting substrate exists, but final immutable result inputs do not.

Interim verified table. It is not a completed section outcome.

Verified resource accounting through the rejected primary pilot (not final)

scope	metric	value	unit	status	source_note
primary_14b_pilot_to_date	actual_allocated_gpu_time	212.281778	seconds	observed	Includes both failed service-start allocations recorded in the cumulative ledger.
primary_14b_pilot_to_date	actual_allocated_gpu_time	0.058967161	hours	observed	Derived exactly from cumulative allocated microseconds for display only.

scope	metric	value	unit	status	source_note
primary_14b_pilot_v2	peak_gpu_vram	10638852096	bytes	observed	Maximum persisted resource sample in the rejected v2 run.
primary_14b_pilot_v2	peak_process_ram	2432729088	bytes	observed	Maximum persisted resource sample in the rejected v2 run.
primary_14b_pilot_v2	peak_project_storage	13623907840	bytes	observed	Maximum persisted resource sample in the rejected v2 run.
study	current_vllm_service_stopped	true	boolean	observed	The rejected primary pilot result records that the service stopped.
study	final_scheduled_gpu_forecast		hours	incomplete	Requires the accepted fallback micro-pilot plus exact development timing block.
study	final_total_runpod_wall_time		hours	incomplete	No final pod-session accounting artifact exists yet.

6. Primary C2 versus C1 results

STATUS: INCOMPLETE — Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.; Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.

No confirmatory or descriptive outcome is available for this section.

This confirmatory section will use twelve worlds as independent units, two one-sided paired t-tests, Holm correction, two-sided paired intervals, all 4096 sign-flip assignments, and the registered bootstrap sensitivity.

7. Required C2 versus C0 results

STATUS: INCOMPLETE — Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.; Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.

No confirmatory or descriptive outcome is available for this section.

C2 versus the credible deterministic C0 baseline is a required secondary comparison on all retained endpoints.

8. Gated C2 versus A-FixedSelect mechanism results

STATUS: INCOMPLETE — Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.; Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.

No confirmatory or descriptive outcome is available for this section.

The mechanism claim is eligible only after the C2-versus-C1 organization gate passes and C2 exceeds mechanically construction-disabled A-FixedSelect on the registered endpoint.

9. Rare-pivotal preservation

STATUS: INCOMPLETE — Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.; Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.

No confirmatory or descriptive outcome is available for this section.

Any favorable claim must accompany rare-pivotal qualified-assertion recall and support-path survival; the registered noninferiority loss margin is 0.05.

10. Entropy and visual-clutter profiles

STATUS: INCOMPLETE — Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.; Visualization and feedback: GPU-backed scripted feedback and researcher traces have not executed.

No confirmatory or descriptive outcome is available for this section.

No entropy direction, node count, crossing count, or cluster count is inherently better. Structural profiles must remain beside fidelity, grounding, validity, and rare-pivotal preservation.

11. Community and cluster organization

STATUS: INCOMPLETE — Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.

No confirmatory or descriptive outcome is available for this section.

The registered panel uses Leiden-CPM with AML, purity, conductance, fragmentation, merging error, descriptive cluster count and modularity, and cross-seed stability.

12. Paraphrase and contrastive behavior

STATUS: INCOMPLETE — Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.; Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.

No confirmatory or descriptive outcome is available for this section.

Contrastive decision-change scoring operates over identical evidence and nonselection ontology changes. Paraphrase stability is interpreted conditional on correctness.

13. Reduced ablation results

STATUS: INCOMPLETE — Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.

No confirmatory or descriptive outcome is available for this section.

Only A-NoContext (12 calls), A-NoTemporalEpistemic (8), and A-NoRareGuard (8) are registered, each with a one-switch-only configuration audit.

14. Feedback and interface demonstrations

STATUS: INCOMPLETE — Visualization and feedback: GPU-backed scripted feedback and researcher traces have not executed.

No confirmatory or descriptive outcome is available for this section.

Six known-answer scripts and three researcher traces demonstrate REFINE_CONTEXT and REQUEST_MERGE_SPLIT. They do not support participant or usability claims.

15. Bounded first-novel transfer study

STATUS: INCOMPLETE — First-novel case study: No lawful local novel path has been supplied and no case-study output exists.

No confirmatory or descriptive outcome is available for this section.

The planned transfer study uses four preregistered windows, eight bounded same-evidence queries, and one separately labeled full-index operational C2 query. Public examples must use paraphrases and opaque evidence IDs.

16. Runtime, GPU, storage, invalid-output, repair, and failure accounting

STATUS: INCOMPLETE — Contracts, resource controls, and GPU pilot: The 14B candidate was rejected and fallback acceptance has not completed.; Results and conference artifacts: The reporting substrate exists, but final immutable result inputs do not.

No confirmatory or descriptive outcome is available for this section.

Failures and invalid or unrepaired outputs remain visible and intention-to-treat. Final accounting is cumulative across every registered base call, repair, failure, timeout, and GPU-service allocation.

Interim verified table. It is not a completed section outcome.

Observed primary-model pilot failures (not final run accounting)

run_id	model_repository	immutable_revision	failure_type	completed_generation_calls	gate_passed	vllm_service_stopped	allocated_gpu_seconds	interpretation
phase1-qwen3-14b-awq-v1	Qwen/Qwen3-14B-AWQ	1a6fe1ecf891437a270cce11ad54d796c4f56ce0	TypeError	0	false	true	0.096330	Rejected primary-model lifecycle start; not a scientific model output.
phase1-qwen3-14b-awq-v2	Qwen/Qwen3-14B-AWQ	1a6fe1ecf891437a270cce11ad54d796c4f56ce0	RuntimeWatchdogTimeout	0	false	true	212.185448	Rejected primary-model lifecycle start; not a scientific model output.

Interim verified table. It is not a completed section outcome.

Verified resource accounting through the rejected primary pilot (not final)

scope	metric	value	unit	status	source_note
primary_14b_pilot_to_date	actual_allocated_gpu_time	212.281778	seconds	observed	Includes both failed service-start allocations recorded in the cumulative ledger.
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17. Error analysis, negative results, and limitations

STATUS: INCOMPLETE — Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.; Metrics, statistics, and ablations: No registered held-out condition table exists to analyze.; First-novel case study: No lawful local novel path has been supplied and no case-study output exists.; Results and conference artifacts: The reporting substrate exists, but final immutable result inputs do not.

No confirmatory or descriptive outcome is available for this section.

The final analysis must retain null and negative findings, all declared error categories, the twelve-world sample limitation, one-model and one-prompt dependence, compute asymmetry, and descriptive-only transfer scope.

18. Exact claim boundaries

STATUS: INCOMPLETE — Results and conference artifacts: The reporting substrate exists, but final immutable result inputs do not.

No confirmatory or descriptive outcome is available for this section.

Claims are conditional on registered gates and confined to the benchmark, pinned model, frozen prompt, and bounded narrative transfer study.

Exact boundaries

- Aggregate 12-world comparisons are confirmatory efficacy evidence; qualitative examples are illustrations only.
- Development examples are pedagogical and cannot support held-out efficacy claims.
- Narrative transfer is descriptive and is not pooled with the synthetic population estimate.
- No result implies general LLM superiority, human usability, complete novel truth, or series-wide generalization.

- Lower entropy, fewer nodes, fewer crossings, and fewer clusters are never inherently better.
- Construction-freedom attribution requires the gated C2 versus A-FixedSelect mechanism result and rare-pivotal safeguards.

19. Easy-to-interpret qualitative illustrations

STATUS: INCOMPLETE — Synthetic benchmark: The mandatory independent condition-blind review has not been supplied or adjudicated.; Conditions and primary comparisons: Development and held-out GPU condition outputs are not complete.; Visualization and feedback: GPU-backed scripted feedback and researcher traces have not executed.; First-novel case study: No lawful local novel path has been supplied and no case-study output exists.

No confirmatory or descriptive outcome is available for this section.

The final report reserves distinct slots for the development tutorial, rare-pivotal fact, temporal or epistemic case, fixed held-out illustration, deterministic failure case, and one or two narrative illustrations. These do not substitute for aggregate efficacy evidence.

Pre-output selection rules

development_tutorial — pedagogical development example: Use syn-dev-01 (easy) and its registered A/B contrast in query-assignment order to explain identical evidence under two contexts and all four condition capabilities. Tie-break: Not applicable: the world and query ordinals are fixed here before output inspection.

rare_pivotal — pedagogical development example: Choose the lexicographically first development world/context whose frozen scorer gold has a positive rare-pivotal denominator and complete support path. Tie-break: Ascending world_id, then context_id, then rare-pivotal assertion anchor.

temporal_epistemic — pedagogical development example: Choose the lexicographically first development context eligible for the frozen A-NoTemporalEpistemic probe, preferring holder-attributed evidence when eligibility ties. Tie-break: Holder-attributed before temporal-only, then ascending world_id and context_id.

held_out_illustration — qualitative held-out illustration: Use the easy world selected by the independent-review seed (syn-test-03), then its context at registered query ordinal A (ctx_556b0577875afcad7419). This selection does not inspect any condition output. Tie-break: The review seed and ordinal-A rule are frozen; no outcome-based tie is permitted.

counterexample — qualitative failure illustration: After intention-to-treat scoring, select the hard-stratum world-context with the minimum mean C2 strict qualified-assertion F1 across the two registered seeds; include invalid and timed-out outputs as registered zeros. Show C0, C1, C2, and A-FixedSelect for the selected unit regardless of which is best. Tie-break: Ascending world_id, then context_id. Within the selected context, display seed blocks in registered order; never pick the worse seed post hoc.

narrative_illustration — descriptive narrative illustration: After the four windows are preregistered and before condition output inspection, use the lowest registered window ordinal and both of its contexts; publish only high-level paraphrases and opaque evidence IDs. Tie-break: Ascending preregistered window ordinal, then registered context ordinal.

Machine-readable provenance

Source manifest SHA-256: 16a65d4bb31bd34c4c0f5564062f348388b4d7d841b1ae80e648e6e0577bc071